Reg. No. | | | | | | | | | | |

[21MSCH3S02]

THIRD SEMESTER UNIVERSITY EXAMINATION, NOVEMBER 2022

ANALYTICAL CHEMISTRY

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. Describe significance testing. How is a significance test constructed?

b. Explain the one-tailed and two-tailed significance test. Write a note on errors in significance testing.

c. Elucidate band broadening in column chromatography using van Deemter equation.

d. Describe the different types of capillary columns used in gas chromatography.

e. Explain the principle of high performance liquid chromatography?

SECTION - B

2. Answer any TWO of the following questions. **2×9=18**

a. Calculate the pH during the titration of 50 ml 0.0500M NaOH with 0.1000M HCl after the addition of **i.** 24.50 ml **ii.** 25 ml and **iii.** 25.5 ml of HCl. Draw the titration curve.

b. Discuss the limitations of classical analytical chemistry in the analysis of biomolecules and how they are solved in bio-analytical chemistry.

c. Explain the 'duck shape' nature of a CV plot for a redox couple involving reversible electron transfer process with a suitable example.

SECTION – C

3. Answer the following question. **12 Marks**

a. i. Splitless injection is used for quantitative analysis and split injection is used for qualitative analysis. Justify this comment.

ii. Create the TG-DTA curves of calcium oxalate monohydrate and discuss the thermogram. Show the calculations for expected weight loss or gains.

iii. Explain the sequential steps you will use to determining the structure of the protein.

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112